

April 12, 2026

Hiring Team
Centre for Immersification
Singapore Institute of Technology
Singapore

Subject: Application for Research Engineer / Research Fellow (AI Agent) - JL

Dear Hiring Team,

I am applying for the Research Engineer / Research Fellow (AI Agent) position at the Centre for Immersification. I have been building game AI systems in Unreal Engine and Unity for the past few years, and your team's work on autonomous agents in immersive 3D environments is exactly the kind of problem I want to spend my time on.

Most of my research deals with AI systems that need to run in real time inside game engines. I have published in *IEEE Access* and *INASS IJIES* on topics like Q-Learning combined with Fuzzy Logic for dynamic difficulty adjustment, and context-aware deep learning for in-game audio. For my thesis at EEPIS, I built *Xanthous*, a VR horror simulation in Unreal Engine where NPCs make decisions through layered Fuzzy Logic, Monte Carlo sampling, and Behavior Trees. Getting those simulations to hold up in VR with tight frame budgets was the hard part, and that kind of optimization problem is what draws me to this role.

Separately, I am working on *Clone World*, where AI agents keep persistent long-term memory and develop their own personalities across play sessions. I built it on Hexagonal Architecture so it stays engine-agnostic, and I am wiring in local LLMs for conversational behavior. The problems I am solving there, like getting agents to act consistently over long sessions and generate believable dialogue, are close to what this role asks for with content moderation and dynamic narratives in Roblox and Minecraft.

On the industry side, I lead a team of 8 at JoyBait Studio and spent 10 months at the Apple Developer Academy shipping iOS and WatchOS apps with CoreML. I write production code in C++, C#, Python, and Swift, and I am used to owning a project from prototype through deployment.

What interests me about SIT is that the research ships. The Centre is building real systems in Roblox and Minecraft with Vision-Language Models and multi-agent setups, and I want to be part of making those work.

I have attached my CV. I would be happy to talk more about any of this. Thank you for your time.

Sincerely,

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